

COMPANION
to

ChatGPT Teaches TikZ

Appendix A: Commands in This Book

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A.1 Using This Reference

This companion collects the commands, operations, keys, options, and syntax patterns listed in the chapter-opening command tables. Each item appears once, under its first substantial introduction. The description is the concise purpose statement used in that chapter, and the final column identifies the chapter in which the item is introduced.

The entries are arranged alphabetically without regard to an initial backslash. Symbolic entries precede the alphabetic entries. This is a reference for recall and navigation; the chapters and their numbered companions provide the explanations, examples, and exercises.

Principle

Use this table to find where an item is introduced. Then return to that chapter when you need context, examples, or the limits of the notation.

A.2 Alphabetical Reference

Command or syntax	Description	In chapter
--	Connects two coordinates with a straight line segment.	1
->	Adds an arrowhead to the end of a path.	1
-	Draws a right-angle connection, moving vertically and then horizontally.	3
;	Ends a TikZ drawing command.	1

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Command or syntax	Description	In chapter
-	Draws a right-angle connection, moving horizontally and then vertically.	3
#1	Refers to the first parameter of a style.	20
#2	Refers to the second parameter of a style.	20
\$(A - B)\$	Uses the x-coordinate of A and the y-coordinate of B.	26
\$(A - B)\$	Uses the x-coordinate of B and the y-coordinate of A.	26
(A)!0.5!(B)	Computes the midpoint of two points.	15
\$(A)!0.5!(B)\$	Computes the midpoint of A and B.	8
\$(A)!d!(B)\$	Selects a point a distance d from A toward B.	26
\$(A)!t!(B)\$	Selects a point a fraction t of the way from A to B.	26
\$(A)!t!(B)\$	Gives the point a fraction t of the way from A to B.	8
\$(A)!t!90:(B)\$	Rotates the direction toward B before locating the point.	26
\$(A)+(B)\$	Adds the coordinate vectors A and B.	8
\$(A)+(v)\$	Adds a displacement to a coordinate.	26
\$(A)+(x,y)\$	Translates A by the vector (x,y).	8
\$(A)-(B)\$	Forms a coordinate difference.	26
\$(A)-(B)\$	Subtracts the coordinate vector B from A.	8
\$(A)-(x,y)\$	Translates A by the opposite vector.	8
a,...,b	Generates a numerical sequence.	23
a,b,c	Supplies an explicit list of loop values.	23
above	Places text above a coordinate.	2
above left=of	Places a node diagonally above and to the left.	6
above=of	Places a node above another node.	6
amplitude=	Controls the height of a decoration.	13

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Command or syntax	Description	In chapter
<code>anchor=</code>	Controls how the contents of a cell are aligned.	12
<code>and</code>	Separates the two Bézier control points.	9
<code>/.append style=</code>	Adds options to the end of an existing style.	19
<code>at</code>	Specifies where a node or coordinate is placed.	2
<code>auto</code>	Lets TikZ choose the label position automatically.	10
<code>background rectangle/.style</code>	Controls that rectangle's appearance.	24
<code>base</code>	Refers to the text baseline of a node.	27
<code>base east, base west</code>	Refers to the ends of the text baseline.	27
<code>\begin{pgfonlayer}</code>	Begins drawing on a selected layer.	24
<code>below</code>	Places text below a coordinate.	2
<code>below right=of</code>	Places a node diagonally below and to the right.	6
<code>below=of</code>	Places a node below another node.	6
<code>bend left</code>	Curves a connection to the left.	3
<code>bend right</code>	Curves a connection to the right.	3
<code>blue!20</code>	Produces a light tint by mixing a color with white.	5
<code>blue!60!black</code>	Mixes two colors in a specified proportion.	5
<code>brace</code>	Draws a decorative brace.	13
<code>Bracket</code>	Bracket arrow tip.	10
<code>bumps</code>	Produces a bumpy path.	13
<code>by=</code>	Assigns names to computed intersection points.	26
<code>/.cd</code>	Changes to a key directory.	21
<code>center</code>	Refers to the center anchor.	27
<code>child</code>	Adds a child node.	7
<code>Circle</code>	Circular arrow tip.	10
<code>circle</code>	Draws a circle.	15
<code>clip</code>	Restricts the visible drawing area.	14
<code>/.code=</code>	Executes TikZ code when a key is used.	21
<code>code=</code>	Contains the drawing commands executed by a <code>pic</code> .	22

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Command or syntax	Description	In chapter
<code>coil</code>	Draws a coiled spring.	13
<code>column sep=</code>	Sets the spacing between columns.	12
<code>construction/.style</code>	Defines a style for auxiliary lines.	18
<code>controls</code>	Specifies Bézier control points.	9
<code>\coordinate</code>	Defines a reusable named point.	2
<code>cycle</code>	Closes the current path.	9
<code>dashed</code>	Draws a path using dashes.	4
<code>decorate</code>	Applies a decoration to a path.	13
<code>decorations.pathmorphing</code>	Loads the decoration library.	13
<code>/.default=</code>	Supplies a value when a style is used without an explicit argument.	20
<code>densely dashed</code>	Draws auxiliary construction lines.	15
<code>dotted</code>	Draws a path using dots.	4
<code>\draw</code>	Draws a path such as a line, curve, or arrow.	1
<code>draw</code>	Draws a border around a node when used as a node option.	2
<code>draw opacity=</code>	Controls outline transparency.	14
<code>draw=</code>	Specifies the outline colour.	18
<code>draw=#1</code>	Uses a parameter to control an outline colour.	20
<code>draw=blue</code>	Sets the outline color of a path or node.	5
<code>east, west</code>	Refers to the right and left anchors.	27
<code>edge</code>	Draws a connection from one node to another.	9
<code>edge from parent</code>	Controls how parent and child are connected.	7
<code>edge/.style</code>	Defines a style for graph edges.	18
<code>\end{pgfonlayer}</code>	Ends drawing on that layer.	24
<code>/.estore in=</code>	Stores an expanded value in a macro.	21
<code>evaluate=</code>	Computes a value while a loop is running.	23
<code>even odd rule</code>	Creates holes in clipped regions.	14
<code>every edge/.style=</code>	Sets defaults for every edge.	19

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Command or syntax	Description	In chapter
<code>every label/.style=</code>	Sets defaults for every label.	19
<code>every node/.style=</code>	Sets defaults for every node in the current scope.	19
<code>every path/.style=</code>	Sets defaults for every path in the current scope.	19
<code>external/export next=</code>	Controls whether the next picture is externalized.	25
<code>external/force remake</code>	Forces externalized pictures to be rebuilt.	25
<code>external/remake next</code>	Forces only the next picture to be rebuilt.	25
<code>fill</code>	Fills a closed region.	14
<code>fill opacity=</code>	Controls fill transparency.	14
<code>fill=</code>	Highlights selected cells.	12
<code>fill=#1</code>	Uses a parameter to control a fill colour.	20
<code>fill=yellow</code>	Fills a closed shape or node with color.	5
<code>filldraw</code>	Draws and fills simultaneously.	14
<code>\filldraw</code>	Marks important points.	15
<code>fit</code>	Creates a node that encloses a collection of nodes or coordinates.	28
<code>fit=</code>	Makes a node enclose specified nodes or coordinates.	24
<code>\foreach</code>	Repeats drawing commands using a loop.	1
<code>grow=down</code>	Makes a tree grow downward.	7
<code>grow=left</code>	Makes a tree grow toward the left.	7
<code>grow=right</code>	Makes a tree grow toward the right.	7
<code>grow=up</code>	Makes a tree grow upward.	7
<code>ifthenelse()</code>	Selects one of two values according to a condition.	23
<code>in=</code>	Specifies the direction in which a curve approaches a point.	9
<code>/.initial=</code>	Gives a key an initial stored value.	21
<code>inner sep=</code>	Controls the space between the text and the node boundary.	27
<code>\input</code>	Reads a separate source file into the current document.	29

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Command or syntax	Description	In chapter
<code>/.is choice</code>	Declares a key whose value is chosen from named alternatives.	21
<code>label/.style</code>	Defines a style for labels.	18
<code>label={#1}</code>	Uses a parameter as visible text.	20
<code>Latex</code>	Traditional arrowhead.	10
<code>left</code>	Places text to the left of a coordinate.	2
<code>left=of</code>	Places a node to the left of another node.	6
<code>let</code>	Stores coordinate or numerical values for later use in a path.	8
<code>level distance</code>	Sets the spacing between successive levels.	7
<code>line width=#1</code>	Uses a parameter to control line thickness.	20
<code>line width=2pt</code>	Sets an exact line width.	4
<code>looseness=</code>	Adjusts the curvature of a path.	9
<code>m-i-j</code>	Refers to the entry in row <i>i</i> and column <i>j</i> .	12
<code>matrix of nodes</code>	Creates a matrix whose entries are nodes.	12
<code>mid, mid east, mid west</code>	Provides anchors based on the middle of the text.	27
<code>minimum height=</code>	Gives a node a minimum vertical size.	27
<code>minimum size=</code>	Gives every cell the same dimensions.	12
<code>minimum size=#1</code>	Uses a parameter to control a dimension.	20
<code>minimum width=</code>	Gives a node a minimum horizontal size.	27
<code>mirror</code>	Reverses the direction of a brace.	13
<code>name</code>	Assigns a name to a vertex.	17
<code>name intersections=</code>	Computes intersections of named paths.	26
<code>name path=</code>	Gives a path a name for later intersection calculations.	26
<code>name prefix=</code>	Adds a prefix to names created inside a <code>pic</code> .	22
<code>name suffix=</code>	Adds a suffix to names created inside a <code>pic</code> .	22
<code>(name)</code>	Gives a coordinate or node an internal name.	2

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Command or syntax	Description	In chapter
<code>\node</code>	Places text or a diagram object in a TikZ picture.	2
<code>node</code>	Labels geometric objects.	15
<code>node distance</code>	Sets the default spacing between positioned nodes.	6
<code>node[midway]</code>	Places a label halfway along a connection.	3
<code>node[near end]</code>	Places a label near the end of a connection.	3
<code>node[near start]</code>	Places a label near the beginning of a connection.	3
<code>node[pos=.25]</code>	Places a label one-quarter of the way along a path.	10
<code>node[pos=.75]</code>	Places a label three-quarters of the way along a path.	10
<code>nodes={\ldots}</code>	Applies the same options to every cell.	12
<code>north east, north west</code>	Refers to the upper corner anchors.	27
<code>north, south</code>	Refers to the top and bottom anchors.	27
<code>on background layer</code>	Places a scope on the background layer.	24
<code>on grid</code>	Aligns positioned nodes using their anchors on the positioning grid.	6
<code>opacity=</code>	Controls the transparency of an object.	14
<code>[options]</code>	Changes the appearance or behavior of a drawing command.	1
<code>out=</code>	Specifies the direction in which a curve leaves a point.	9
<code>parabola</code>	Draws a parabolic arc.	9
<code>\path</code>	Creates nodes and coordinates without necessarily drawing a visible path.	28
<code>\pgfdeclarelayer</code>	Declares a named drawing layer.	24
<code>\pgfkeysvalueof</code>	Retrieves the stored value of a key.	21
<code>\pgfmathparse</code>	Evaluates a PGF mathematical expression.	23
<code>\pgfmathresult</code>	Contains the result of the most recent parse.	23
<code>\pgfmathsetmacro</code>	Computes a value and stores the result in a macro.	23

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Command or syntax	Description	In chapter
<code>\pgfmathtruncatemacro</code>	Computes and stores an integer result.	23
<code>\pgfsetlayers</code>	Specifies the order of the layers.	24
<code>\pic</code>	Places a previously defined <code>pic</code> .	22
<code>pic</code>	Places a reusable graphical component.	28
<code>pic actions</code>	Applies options supplied to the <code>pic</code> 's drawing actions.	22
<code>pic text</code>	Stores text supplied to a <code>pic</code> .	22
<code>pic text options</code>	Styles text associated with a <code>pic</code> .	22
<code>pics/name/.style=</code>	Defines a reusable <code>pic</code> .	22
<code>positioning</code>	Name of the TikZ positioning library.	6
<code>/.prefix style=</code>	Adds options to the beginning of an existing style.	19
<code>prefix=</code>	Selects the folder used for externalized graphics.	25
<code>right</code>	Places text to the right of a coordinate.	2
<code>right=2cm of</code>	Places a node a specified distance to the right.	6
<code>right=of</code>	Places a node to the right of another node.	6
<code>rotate around={...}</code>	Rotates about a specified point.	11
<code>rotate=</code>	Rotates a drawing.	11
<code>rounded corners</code>	Rounds corners of rectangles and polygonal paths.	5
<code>rounded corners=...</code>	Sets the amount of rounding.	5
<code>row sep=</code>	Sets the spacing between rows.	12
<code>scale=</code>	Enlarges or reduces a drawing.	11
<code>scope</code>	Limits the effect of transformations.	11
<code>segment length=</code>	Controls the size of a decoration.	13
<code>-shell-escape</code>	Allows L ^A T _E X to launch the auxiliary compilation processes required by the usual externalization workflow.	25

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Command or syntax	Description	In chapter
<code>shift=</code>	Moves a <code>pic</code> by a specified vector.	22
<code>shift={ (x,y) }</code>	Translates a drawing.	11
<code>show background rectangle</code>	Draws a rectangle behind a picture.	24
<code>sibling distance</code>	Sets the spacing between neighboring children.	7
<code>sin and cos</code>	Draw sinusoidal path segments.	9
<code>sloped</code>	Rotates a label to follow the path.	10
<code>snake</code>	Draws a snake-like path.	13
<code>south east, south west</code>	Refers to the lower corner anchors.	27
<code>Stealth</code>	Modern arrowhead.	10
<code>-{Stealth}</code>	Draws a connection with a modern arrowhead.	3
<code>/.store in=</code>	Stores the value of a key in a macro.	21
<code>.style</code>	Defines a reusable collection of options.	2
<code>/.style</code>	Collects options into a reusable style.	28
<code>style A,style B</code>	Combines previously defined styles.	19
<code>/.style args</code>	Defines a style whose appearance depends on parameters.	28
<code>/.style args=</code>	Defines a style whose arguments follow a specified pattern.	20
<code>[style-name]</code>	Applies a previously defined style.	2
<code>/.style=</code>	Defines a reusable style.	18
<code>/.style={\ldots}</code>	Defines a style that may contain a parameter.	20
<code>swap</code>	Places a label on the opposite side of a path.	10
<code>text</code>	Holds the visible text printed by a node.	2
<code>text opacity=</code>	Controls text transparency.	14
<code>thick</code>	Makes a path thicker than the default.	4
<code>thin</code>	Draws auxiliary segments.	16
<code>/tikz/\ldots</code>	Places a key in TikZ's key directory.	21
<code>\tikzexternaldisable</code>	Temporarily turns externalization off.	25

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Command or syntax	Description	In chapter
<code>\tikzexternalenable</code>	Turns externalization on again.	25
<code>\tikzexternalize</code>	Turns externalization on.	25
<code>tikzpicture</code>	The environment that begins and ends a TikZ picture.	1
<code>\tikzset</code>	Defines the visual vocabulary of the diagram.	28
<code>\tikzsetnextfilename</code>	Gives the next externalized picture a stable file name.	25
<code>\tikzset{...}</code>	Defines TikZ styles and settings.	19
<code>to</code>	Creates a flexible connection that can include bends and labels.	3
<code>transform shape</code>	Applies transformations to node shapes.	11
<code>\usetikzlibrary</code>	Loads TikZ libraries required by the project.	29
<code>\usetikzlibrary{intersections}</code>	Loads path-intersection calculations.	26
<code>\usetikzlibrary{arrows.meta}</code>	Loads the modern arrow library.	10
<code>\usetikzlibrary{backgrounds}</code>	Loads convenient background-layer tools.	24
<code>\usetikzlibrary{calc}</code>	Loads the coordinate-calculation library.	8
<code>\usetikzlibrary{external}</code>	Loads the TikZ externalization library.	25
<code>\usetikzlibrary{fit}</code>	Loads nodes that fit around other objects.	24
<code>\usetikzlibrary{matrix}</code>	Loads the matrix library.	12
<code>vertex/.style</code>	Defines a style for graph vertices.	18
<code>very thick</code>	Makes a path more strongly emphasized.	4
<code>(x,y)</code>	Specifies a point in the coordinate plane.	1
<code>\x/\y</code>	Assigns several loop variables from paired data.	23
<code>xscale=</code>	Horizontal scaling.	11
<code>xshift=</code>	Horizontal translation.	11
<code>yscale=</code>	Vertical scaling.	11
<code>yshift=</code>	Vertical translation.	11
<code>zigzag</code>	Draws a zigzag path.	13